

Basics 44: Guiding-Center Theory

VI - Kinetic theory and energetic particles

Separation of gyromotion from slow orbit dynamics, adiabatic invariants, mirror force, and reduced equations of motion

Textbook draft, version 1.0. A fusion alpha particle can execute billions of gyro-orbits before its macroscopic confinement changes. Guiding-center theory removes that rapidly rotating phase while preserving the slow motion needed for orbit following, resonances, and wall-loss calculations. This chapter develops the ordering, magnetic moment, mirror force, principal drifts, validity conditions, and numerical implications of the reduction.

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1 Learning objectives

After completing this note, the reader should be able to:

- Identify the small parameter that justifies gyro-averaging.
- Separate particle position into guiding-center and gyroradius contributions.
- Derive and interpret the first adiabatic invariant μ .
- Derive the mirror force and trapped-particle condition qualitatively.
- State first-order guiding-center equations and principal drifts.
- Recognize when guiding-center theory fails.

2 Prerequisites and reading strategy

Recommended prerequisites are Basics 2, 13, 41, and 42. Review uniform-field gyromotion before studying nonuniform-field corrections.

3 Separation of scales

In a magnetic field, a charged particle rotates at gyrofrequency

$$\Omega_c = \frac{|q|B}{m} \quad (3.1)$$

with Larmor radius

$$\rho_L = \frac{v_\perp}{\Omega_c}. \quad (3.2)$$

If the equilibrium varies over length L_B and time τ_B , guiding-center theory requires

$$\epsilon = \frac{\rho_L}{L_B} \ll 1, \quad \frac{1}{\Omega_c \tau_B} \ll 1. \quad (3.3)$$

The particle position is decomposed as

$$\mathbf{r} = \mathbf{R} + \boldsymbol{\rho}, \quad (3.4)$$

where $\mathbf{R}$ is the slowly moving guiding center and $\boldsymbol{\rho}$ is the rapidly rotating gyroradius vector.

4 Uniform-field orbit

For $\mathbf{E} = 0$ and uniform $\mathbf{B} = B\mathbf{b}$,

$$m\dot{\mathbf{v}} = q\mathbf{v} \times \mathbf{B}. \quad (4.1)$$

The parallel velocity is constant, while the perpendicular velocity rotates. The kinetic energy and pitch components satisfy

$$\frac{mv^2}{2} = \frac{mv_\parallel^2}{2} + \frac{mv_\perp^2}{2}. \quad (4.2)$$

This exact solution is the zeroth-order motion about which the asymptotic expansion is built.

5 Magnetic moment as an adiabatic invariant

Define

$$\boxed{\mu = \frac{mv_\perp^2}{2B}}. \quad (5.1)$$

The perpendicular kinetic energy is $E_\perp = \mu B$. When B changes little during one gyro-orbit, gyro-averaging shows

$$\frac{d\mu}{dt} = O(\epsilon^2) \quad (5.2)$$

for slowly varying static fields. Thus μ is approximately conserved.

A useful physical derivation treats the gyro-orbit as a current loop. Its magnetic moment is

$$I\mathcal{A} = \frac{q\Omega_c}{2\pi}\pi\rho_L^2 = \frac{mv_\perp^2}{2B}\text{sgn}(q). \quad (5.3)$$

The orbital magnetic dipole interacts with field gradients, producing a force proportional to $-\mu\nabla B$.

6 Mirror force

For static fields and negligible electric work, the total kinetic energy

$$\mathcal{E} = \frac{mv_{\parallel}^2}{2} + \mu B \quad (6.1)$$

is constant. Differentiate along the orbit:

$$mv_{\parallel}\dot{v}_{\parallel} + \mu v_{\parallel} \mathbf{b} \cdot \nabla B = 0. \quad (6.2)$$

For $v_{\parallel} \neq 0$,

$$\boxed{m\dot{v}_{\parallel} = -\mu \nabla_{\parallel} B}. \quad (6.3)$$

A particle moving into larger B converts parallel energy into perpendicular energy until $v_{\parallel} = 0$, where it reflects.

If the particle begins where the field is B_0 with pitch angle α_0 , reflection occurs when

$$\sin^2 \alpha_0 \geq \frac{B_0}{B_{\max}}. \quad (6.4)$$

This divides trapped and passing populations.

7 Guiding-center drifts

The lowest-order guiding-center velocity is parallel streaming plus the electric drift:

$$\dot{\mathbf{R}} = v_{\parallel} \mathbf{b} + \mathbf{v}_E + \mathbf{v}_{\nabla B} + \mathbf{v}_c + \dots \quad (7.1)$$

The principal drifts are

$$\mathbf{v}_E = \frac{\mathbf{E} \times \mathbf{B}}{B^2}, \quad (7.2)$$

$$\mathbf{v}_{\nabla B} = \frac{\mu}{qB} \mathbf{b} \times \nabla B, \quad (7.3)$$

$$\mathbf{v}_c = \frac{mv_{\parallel}^2}{qB} \mathbf{b} \times \boldsymbol{\kappa}, \quad \boldsymbol{\kappa} = \mathbf{b} \cdot \nabla \mathbf{b}. \quad (7.4)$$

The $\mathbf{E} \times \mathbf{B}$ drift is independent of particle mass and charge. Grad- B and curvature drifts reverse with charge and grow with particle energy, making them central to fast-ion confinement.

A consistent first-order parallel equation is

$$\boxed{\dot{v}_{\parallel} = \frac{q}{m} E_{\parallel} - \frac{\mu}{m} \mathbf{b} \cdot \nabla B + \dots}. \quad (7.5)$$

8 Guiding-center coordinates and measure

A common coordinate set is

$$(\mathbf{R}, v_{\parallel}, \mu, \vartheta_g), \quad (8.1)$$

where ϑ_g is gyroangle. Gyro-averaged models remove explicit dependence on ϑ_g . The phase-space measure is not simply $d^3R dv_{\parallel} d\mu$; a magnetic Jacobian factor appears. Structure-preserving formulations use a modified field $B_{\parallel}^*$ and measure proportional to

$$B_{\parallel}^* d^3R dv_{\parallel} d\mu d\vartheta_g. \quad (8.2)$$

Ignoring this factor can violate particle conservation in noncanonical coordinates.

9 Orbit frequencies and resonance

A guiding-center orbit can contain fast parallel transit or bounce motion and slower toroidal precession. These generate frequencies

$$\omega_t, \quad \omega_b, \quad \omega_d, \quad (9.1)$$

which enter Alfvén-wave resonance conditions. Module 3 must compute these consistently with the same equilibrium and coordinate conventions used by Module 4.

10 Energy and invariant checks

In time-independent electrostatic fields, an ideal guiding-center integrator should preserve

$$\mathcal{H} = \frac{mv_{\parallel}^2}{2} + \mu B + q\phi \quad (10.1)$$

to numerical accuracy. In an axisymmetric field, canonical toroidal momentum provides another benchmark. Monitoring only spatial trajectories can hide large invariant errors.

11 Limits of validity

Guiding-center theory can fail when:

- ρ_L/L_B is not small;
- fields vary near the gyrofrequency;

- the orbit passes close to magnetic nulls where $B \rightarrow 0$;
- sharp material boundaries require full-orbit impact geometry;
- cyclotron resonances or finite-Larmor-radius effects are essential.

Near marginal validity, a full-orbit comparison is required.

12 Numerical considerations

Runge–Kutta integration is simple but can accumulate secular invariant drift. Symplectic, variational, or volume-preserving methods better respect long-time orbit structure. Adaptive stepping must resolve bounce points and rapid field variation without introducing biased loss boundaries.

13 Connection to the Kinetic MHD-PINN project

Module 3 uses guiding-center trajectories to estimate confinement, orbit frequencies, prompt losses, and wall strikes. Module 5 uses orbit frequencies and mode overlap for resonant drive. Structure-preserving neural surrogates may accelerate orbit maps, but they must be tested against energy, magnetic-moment, and phase-space-volume constraints.

14 Computational laboratory

1. Integrate full Lorentz motion in a uniform field and recover Ω_c and ρ_L .
2. Add a weak field gradient and compare the measured grad- B drift with theory.
3. Integrate the reduced mirror equations in $B(z) = B_0(1 + z^2/L^2)$.
4. Verify conservation of μ and energy as the scale ratio ρ_L/L is reduced.
5. Compare ordinary Runge–Kutta and a structure-preserving method over many bounce periods.

15 Summary

Guiding-center theory exploits the small ratio of gyroradius to equilibrium scale. It removes fast gyroangle dynamics while retaining parallel streaming, mirror motion, and cross-field drifts. The magnetic moment is the key first adiabatic invariant. The reduction enables practical energetic-particle orbit following, but its Jacobian, invariants, and validity limits must be respected.

16 Exercises

1. Estimate the gyroradius of a 3.5 MeV alpha particle in a 5 T field.
2. Derive the mirror reflection condition from energy and μ conservation.
3. Show dimensionally that each drift velocity has units of speed.
4. Explain why $\mathbf{E} \times \mathbf{B}$ drift is independent of charge sign.
5. Propose invariant-based acceptance tests for a Module 3 orbit solver.

Symbols and acronyms used in this document

Symbol / acronym	Name	Meaning in this document
Ω_c	Gyrofrequency	Angular frequency of cyclotron motion.
ρ_L	Larmor radius	Radius of the fast gyromotion.
μ	Magnetic moment	First adiabatic invariant $mv_\perp^2/(2B)$.
$\mathbf{R}$	Guiding-center position	Gyro-averaged particle position.
κ	Field-line curvature	$\mathbf{b} \cdot \nabla \mathbf{b}$.

17 References

1. R. G. Littlejohn, guiding-center Hamiltonian theory papers.
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4. A. J. Brizard and T. S. Hahm, “Foundations of nonlinear gyrokinetic theory,” *Reviews of Modern Physics*, 2007.